# Syed Omer Ali

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## Summary

Security-focused machine learning researcher and software engineer with 3+ years of experience securing and modernizing production web applications. Research covers threat detection, on-device machine learning, model compression, and predictive monitoring of computer-controlled systems.

## Core Skills

- Security: vulnerability assessment, vulnerability remediation, secure coding, threat detection, phishing detection, malware and ransomware analysis, Remote Access Trojan (RAT) detection, threat intelligence integration, Android security, SHA-256 hashing, cryptographic verification
- Machine Learning: deep learning, convolutional neural networks (CNN), VGG16, transfer learning, data augmentation, model compression and optimization, on-device machine learning, PyTorch, OpenCV, natural language processing
- Programming: Python, JavaScript, PHP, C++
- Web and Backend: Node.js, Django, Laravel, React, REST API integration, HTML5, CSS3
- Databases: MySQL, PostgreSQL, MongoDB
- Systems: server deployment, cPanel, production debugging, software development life cycle (SDLC)

## Education

Avila University, Kansas City, MO Master of Science, Computer/Information Technology Services Administration and Management | Expected December 2026 | GPA 3.5/4.0

Mahaveer Institute of Science and Technology, Hyderabad, India Bachelor of Technology, Computer Science Engineering | August 2022

Quli Qutub Shah Diploma/Polytechnic College, Hyderabad, India Diploma, Computer Science Engineering | May 2019

## Experience

Capstone Research: Digital Twins for Predictive Maintenance of Computer-Controlled Systems | Avila University | 2026, In Progress

- Investigating how digital twin technology can improve predictive maintenance through real-time monitoring and early detection of potential system failures.
- Evaluating how digital twin-based maintenance can reduce system downtime and maintenance costs while improving reliability and performance.

Graduate Researcher | Avila University | January 2024 to Present

- Designed lightweight CNN and VGG16-based architectures, reducing model size by 23% and inference latency by 18%.
- Implemented selective layer fine-tuning in PyTorch, reducing on-device adaptation time from 22.

Undergraduate Researcher | Mahaveer Institute of Science and Technology | January 2021 to December 2022

- Built a human activity recognition model using CNN/VGG16 in Python and PyTorch, reaching 96.95% accuracy.
- Applied transfer learning and OpenCV-based feature extraction for real-time action recognition; designed the end-to-end pipeline from video frames to classification.
- Led a team developing CNN models trained on the Kinetics action recognition dataset.
- Combined RFID signal patterns with machine learning algorithms to improve motion interpretation.

## Professional Experience

Software Developer to Tech Team Lead | Accentel Software Solutions | Hyderabad, India | August 2022 to August 2024

- Identified security vulnerabilities in 15+ legacy PHP websites and upgraded them to current, supported versions. [Add vulnerability types and outcomes.]
- Developed and maintained backend systems, database workflows, and server-side logic for large-scale applications.
- Implemented monitoring systems for iOS and Android logistics and network field applications.
- Led client interactions, wrote SRS documentation, and translated requirements into technical specifications.
- Mentored and supervised a developer team, conducted code reviews, and ensured delivery quality and deadline adherence.
- Managed production environments, cPanel operations, deployments, and live server maintenance.

Process Associate | HRH Pvt. Ltd | Hyderabad, India | September 2020 to August 2022

- Managed operational workflows and data processes on OneView, Workforce, and Freshdesk.
- Maintained organizational profiles and internal databases; defined KPIs and simplified workflows with clients.
- Supported production operations, including server deployment, monitoring, and backups.

Web Developer | Creatick Solutions | Hyderabad, India | January 2018 to December 2019

- Developed and maintained responsive web applications using HTML, CSS, JavaScript, jQuery, React.js, Angular, and Vue.js.
- Designed web layouts and prototypes using Figma and CodePen; managed domain configuration and client-side deployment.

## Mentoring and Teaching

- Mentored and supervised a developer team as Tech Team Lead at Accentel Software Solutions, guiding junior developers on code quality, delivery, and deadlines.
- Conducted code reviews to improve the reliability, clarity, and efficiency of team code.
- Led a team developing CNN models for human action recognition during undergraduate research.

## Projects

Phishing Detection System | Python | 2022

- Built a system using NLP and machine learning classifiers to analyze email content, links, and attachment metadata for phishing and social engineering indicators.
- Integrated rule-based validation with third-party threat intelligence checks, and designed the system to update continuously as attack signatures change.
- Wrote detection scripts for malicious permission behavior, including ransomware patterns such as WannaCry.

Android Remote Access Trojan (RAT) Detection | 2020

- Analyzed RAT infections on Android devices below version 9 and built a prototype that detects malicious permission takeover and force-revokes unauthorized admin access.
- Presented findings to faculty.

SHA-256 Secure Online Voting System | Hackathon Winner | 2021

- Designed a voting prototype using SHA-256 hashing to verify votes and prevent tampering.
- Designed vote signature generation, validation logic, and the backend security flow.

QQ Student Portal | Android | 2018

- Led development of an Android-based student portal for results, lab tasks, and service requests, delivered to the department head within two months and used by 300+ students.

## Awards and Honors

- Optimal System Solution Award, OSPS, 2023
- IoT (Raspberry Pi) Human Activity Recognition Project Winner, PanTech, 2021
- SHA-256 Secure Voting System Hackathon Winner, Badruka Engineering College, 2021
- Excelerate Hackathon Winner, JNTUH, 2020

## Certifications

- Cybersecurity Ethics Certification, Udemy, 2020
- Python Certification, Progate, 2022
- Industrial Training Certification, Creatick Solutions, 2019